

HANY EL ATLASSI

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Profile

AI & Security engineer with hands-on experience building and deploying ML systems on AWS: voice pipelines, multi-agent orchestration, federated learning, and the security infrastructure underneath all of it, ensuring complete deployment. Available and open to relocation.

Work Experience

Axon | *Founder*

2026 – Present

- Founding an **agentless, AI-native security posture management** platform that continuously discovers, analyses, and secures the AI infrastructure organisations run: models, data, **MCP** servers, agent architectures, and the cloud/data posture beneath, surfacing vulnerabilities and misconfigurations before and during deployment
- Built to align with emerging standards (**ISO 42001**, NIST AI RMF, OWASP); pre-seed, building the MVP within **HackNation's Venture Lab** (out of MIT)

Attijari Payment (APP) | *AI/ML Engineering Intern*

Jul 2026 – Sep 2026

- Built an **on-premise** AI agent that triages incident tickets in card-payment operations (monétique) by *verified* urgency (P1–P4): a **local** LLM (qwen3:8b via Ollama) investigates the operational database read-only, isolates root cause, and drafts a diagnosis and resolution plan, human-in-the-loop validation, **zero external API calls** (PCI-DSS constraint)
- Tool-driven investigation (read-only SQL guard, BM25 retrieval, full audit trail) lifted triage accuracy to **94%** vs **65%** for the same model with no tools; locked down by a read-only DB role and **36** automated security tests

3D Smart Factory | *Machine Learning Intern*

Jun 2024 – Sep 2024

- Implemented Superpoint Transformer and trained on **30 GB+** Stanford 3D Indoor Scenes, reaching **90%** accuracy and **70%** mean IoU on 3D point cloud segmentation; end-to-end from raw data preprocessing to a deployed real-time Streamlit demo

CIAM AI Club, ENSAM | *President* (200+ members)

Sep 2023 – Jun 2025

- Grew and ran ENSAM's AI community to **200+** members; organised weekly technical workshops, and coordinated student participation in national AI competitions

Education

École Nationale Supérieure d'Arts et Métiers

2022 – 2027

Engineer's Degree — Cybersecurity & Cloud Engineering

GPA: 3.7/4.0

- Research paper in preparation: FedLoRA-MAML, federated second-order meta-learning for personalised accented speech recognition (Wav2Vec2, LoRA, Per-FedAvg, L2-ARCTIC corpus)

Projects

InterviewForge — Voice AI Interview Platform

2026 – Present

- Designing and building a **5-layer concurrent voice architecture**: Deepgram Flux + Pipecat Smart Turn v3 (**12 ms** CPU) → gpt-4o-realtime-mini filler at **~329 ms** → Cartesia Sonic-3 (**40–90 ms** TTFB) over LiveKit WebRTC.
- **7-agent LangGraph** pipeline; knowledge grounded via Claude cached prefix (**90%** cost reduction vs RAG); Judge0 + gVisor sandboxed code execution (seccomp, cgroups, zero network egress); K8s deployment behind Kong and Cloudflare WAF

VoiceFL-MAML — Federated Meta-Learning for Accented Speech | github

2026

- Designed a controlled **4-condition** study (pretrained, centralised fine-tuning, FedAvg, FedLoRA-MAML) on AfriSpeech-200 (**375** speakers, **5** African accent groups), scored with **95%** bootstrap CIs across **77** held-out meta-test speakers
- Engineered a custom differentiable CTC loss in native PyTorch to unlock true second-order MAML; nn.CTCLoss has no registered second derivative, this was the part that made the math actually work
- Cut per-round communication **~295×** by transmitting only **320K** LoRA parameters (rank-16 adapters on transformer layers 6–11) instead of 94.5M full encoder weights

Achievements

- **1st Place, AgorAI Hackathon** — won with Aegis, an AI policy impact simulator for Moroccan AI regulation; presented at UM6P AI for Impact alongside Yann LeCun, Eric Xing, researchers from Google DeepMind, and universities worldwide
- **2nd Place, HackAI 5th Edition (1337AI)** — won with best pitch award, building a voice-first AI agronomist for Moroccan farmers; full farming plan orchestration, hallucination-hardened via a 5-layer safety pipeline, responding in Darija
- **Top 10 globally, Mistral AI Hackathon** — AI clinical safety assistant built on Mistral OCR + real-time voice pipeline
- **CTF: Top 3%** worldwide on HackTheBox; top **10–15** nationally across multiple Moroccan CTF competitions

Skills & Certifications

- **AI/ML:** PyTorch, TensorFlow, Scikit-Learn, LangGraph, LangChain, Flower (Federated Learning), Computer Vision, NLP
- **Cloud & DevOps:** AWS, Kubernetes, Docker, Terraform, ArgoCD, Git/GitHub, Linux
- **Security:** OS/memory forensics, network analysis, NTFS artifacts, prompt injection hardening
- **Languages:** Python, C/C++, Java, SQL, Bash | English (Fluent), French (Fluent), Arabic (Native)
- **Certifications:** IBM AI Engineering, Deep Learning and Reinforcement Learning Specializations